

Bayesian Prior Persistence

Trust the market where it speaks, the prior where it does not

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Abstract

A desk does not re-derive its volatility surface from scratch each morning: it *persists* yesterday's surface and lets today's quotes update it. Done naively this is a dilemma. Load the prior too hard and you *damp* a real overnight signal — an ATM vol that genuinely jumped gets pulled back toward yesterday. Load it too softly and the sparsely-quoted wings *flap*, over-fit to a steep ATM skew. This note documents the Vol-Fitter's resolution: a Bayesian prior-persistence framework built on one idea — an *activation gate* that turns the prior *off* exactly where today's quotes already identify a feature, and *on* where they do not. We give the universal residual form and the gate gap = $\max(1 - \text{obs}/\text{req}, 0)^\gamma$, the shared precision vocabulary (quote density, bid-ask width, freshness, transport distance), and the seven persistence modes — off, overlay, strike gaps, quote operators, smile factors, hybrid (the default), and graph-only — including why the risk-reversal/butterfly operators must be *signed baskets* to preserve their coupling. A worked example, built with the production calibrator, shows an ATM jump of +8 vol points preserved (the persisted ATM is 40.7%, tracking the market, not the calmer prior's 31.5%) while an unquoted wing is held at the prior (40.6% vs the market-only fit's wild 52.6%). An empirical temporal backtest confirms the hybrid default reconstructs held-out wings ~ 32 vol bps better than no prior, two-thirds of the time.

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1 The morning dilemma

Overnight, two things change unevenly. The at-the-money level may move sharply on fresh news, and it is *well quoted* — the liquid near-the-money options re-price immediately. The wings, by contrast, may have *no* fresh quotes at all, yet they still need values. A fitter faces a choice:

- **Re-fit from today’s quotes only.** The ATM is captured, but with no wing quotes the model extrapolates the (now steeper) ATM skew into the wings, producing implausible tails — the wings *flap*.
- **Anchor hard to yesterday’s prior.** The wings are stabilized, but the prior also pulls the *ATM* back toward yesterday, *damping* the real overnight signal.

Heuristic

The resolution is to make the prior’s strength *local* and *data-aware*: strong in the wings (no quotes $\Rightarrow$ trust yesterday), zero at the money (fresh quotes $\Rightarrow$ trust today). The same prior, applied through a gate that reads how well today’s data identifies each feature, gives both behaviours at once. This is just Bayesian updating with a precision that depends on the observation density — but stated as an operational rule: *do not damp the signal*.

2 The universal residual and the activation gate

Every persistence mode adds residuals of one shape: a difference between a *functional* of the model and the same functional of the prior, scaled by a data-aware weight,

$$r_j = \sqrt{\lambda_j} \frac{O_j(\text{model}) - O_j(\text{prior})}{\text{scale}_j}, \quad \lambda_j = \underbrace{\text{strength}}_{\text{global}} \cdot \underbrace{\pi_j^{\text{prior}}}_{\text{prior precision}} \cdot \underbrace{\text{gap}_j}_{\text{gate}}. \quad (1)$$

O_j is a strike price, an operator (ATM/RR/BF), or a local smile factor depending on the mode (section 4). The decisive factor is the *activation gate*

$$\text{gap}_j = \max\left(1 - \frac{\pi_j^{\text{obs}}}{\pi_j^{\text{req}}}, 0\right)^\gamma, \quad (2)$$

where π_j^{obs} is how precisely today’s quotes identify coordinate j and π_j^{req} the precision at which the prior should switch off. When the data meets or exceeds the requirement, $\text{gap}_j = 0$ and the prior contributes *nothing*; when the data says nothing, $\text{gap}_j \rightarrow 1$ and the prior is at full strength. The exponent γ sharpens the transition. Figure 1 plots it. The gate and the prior precision it produces are two lines (matches production to 10^{-13}):

Listing 1: The universal activation gate (2) and the gated prior precision (distilled from `calib/precision.py`).

```
1 def activation_gap(obs, req, gamma=1.0):      # 1 where data says
    nothing, 0 where it identifies
2     return np.clip(1 - obs/np.maximum(req, 1e-12), 0.0, 1.0)**gamma
3
```

```

4 def active_prior_precision(base, gap):          # prior precision
    actually entering the fit
5     return base * gap                          # ~0 where the data is
    already sufficient

```

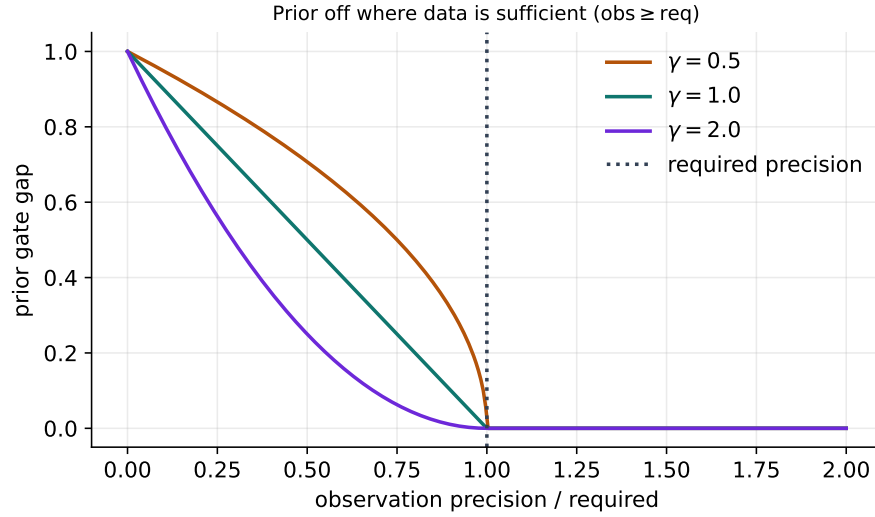


Figure 1: The activation gate (2). Past the required precision the prior weight is exactly zero — the “do not damp the signal” rule. γ controls how hard the on/off edge is.

3 The precision vocabulary

The gate needs a shared notion of precision, used identically by every mode (and by the graph baseline of Note 14). A raw precision is scaled by model-agnostic factors in $(0, 1]$:

- **Fit quality** $1/\text{rms}^2$, floored at a 1-vol-bp excellence level (a great fit is precise, not infinitely so).
- **Quote density** — more supporting quotes raise precision, saturating at a reference count (~ 8 near-ATM quotes for full credit).
- **Bid–ask width** — precision halves at a relative spread of $\sim 5\%$ of ATM vol.
- **Freshness** — exponential decay with quote/prior age (half-lives of a few days for observations, ~ 30 days for the prior).
- **Transport distance** — a prior transported a forward distance $|h| = |\log(F_{\text{now}}/F_{\text{prior}})|$ is less trustworthy, decaying on a scale of $\sim 10\%$.

The *observation* precision (data side) combines fit quality, density, spread and freshness; the *prior* precision (baseline side) combines provenance, age and transport. Both carry per-handle floors and caps so that one excellent fit cannot dominate and one stale prior cannot vanish entirely.

4 The seven modes

The mode (`priorPersistenceMode`) chooses the functional O_j in (1).

1. **Off.** No prior penalty and no overlay — a pure current-market fit.
2. **Overlay.** The transported prior is *drawn* (dotted) for reference but exerts no calibration pull.
3. **Strike gaps.** O_j is the call price at a spread of delta-locations. The weight follows the *data-gap* density: $\rho_{\text{desired}}(x) - \rho_{\text{observed}}(x)$, the positive part, where ρ_{observed} is a Gaussian kernel density of the live quotes and ρ_{desired} is the coverage we want spread over the *wider* delta span. A region already sampled to the desired density gets zero prior weight; only the deficit (sparse wings, gaps) is filled. This is the mode in the worked example.
4. **Quote operators.** $O_j \in \{\text{ATM}, \text{RR}_{25}, \text{BF}_{25}, \text{VarSwap}\}$, persisted only where today’s quotes do not identify the operator. RR and BF must be *signed baskets* (section 4.1).
5. **Smile factors.** $O_j \in \{\text{ATM}, \text{skew}, \text{curvature}, \dots\}$, ATM-local finite-difference stencils — the prior’s *shape* re-expressed at the current node.
6. **Hybrid (default).** Operator priors (ATM/RR/BF/var-swap) plus a residual deep-tail strike anchor only where no operator covers — the empirically best reconstruction (section 7).
7. **Graph-only.** The prior enters *only* as the graph baseline for extrapolation (Note 14); dark nodes receive propagated innovations rather than a direct anchor.

4.1 Why operators are signed baskets

A risk-reversal or butterfly is a *combination* of leg vols, not a single quote. Persisting RR and BF as *per-leg* prices would silently drop their coupling — the prior would anchor the legs independently and lose “the skew should persist” as a joint statement. Instead each operator is a signed linear functional (basket) of the leg implied vols (parametric) or of the leg call prices (local-vol), giving *one* residual per operator that preserves the RR/BF coupling on whatever surface the model draws. This also fixed a long-standing asymmetry: the SVI and Multi-Core SIV overlays, which previously received no prior at all, now get the same signed-operator treatment as LQD.

5 Two-pass activation

The gate (2) needs the observation precision, which in its sharpest form is the *realized* fit quality — a chicken-and-egg with the fit it modifies. The optional two-pass mode (`priorDataOnlyPrepass`) resolves it cleanly: fit once on data alone, measure each operator’s realized precision, then refit with priors active only on the under-observed coordinates. This is the most faithful “do not damp” behaviour but costs $\sim 2\times$ per node, so it is opt-in; the default single pass gates by quote-support precision with no extra fit.

6 Worked example: an overnight jump

Figure 2 is the dilemma made concrete, built with the production strike-gap prior and calibrator. Yesterday’s prior is a calm 31.5%-ATM smile. Overnight the market jumps: today’s ATM is 39.9% with a steeper skew, and *only* the ATM region ($|k| \leq 0.10$) is quoted. The market-only fit captures the ATM but extrapolates the steep skew into a wild put wing (52.6% at $k = -0.30$). The prior-persisted fit keeps the ATM essentially at the *market* level (40.7%, tracking the market’s 39.9%

rather than being damped toward the prior’s 31.5%) because the gate is zero where the quotes are dense, while holding the unquoted wing at the prior (40.6%, vs the prior’s 41.7%) because the gate is one where there are no quotes. The signal is preserved; the gap is filled.

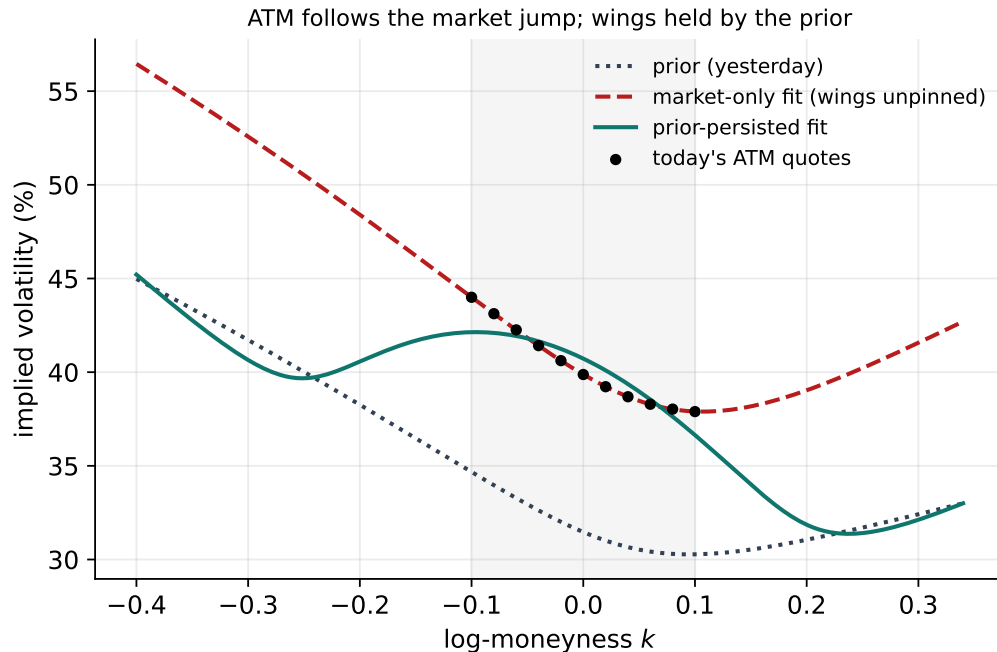


Figure 2: “Don’t damp the signal.” The persisted fit (solid) tracks the market ATM jump (quoted region shaded) while the unquoted wings are held by the prior (dotted), rather than flapping like the market-only fit (dashed).

7 Empirical validation

The framework is validated by a *temporal* backtest. For each (asset, $T-1 \rightarrow T$) pair it fits day $T-1$ ’s full chain, freezes it as the active prior, thins day T to its ATM region, refits under each mode, and scores the reconstructed *held-out* moderate wing against the true day- T quotes (with *off* as the baseline). Over the full spike regime (1117 nodes plus a bandwidth/probe sweep), the shipped **hybrid** default reconstructs the held-out wing ~ 32 vol bps better than no prior, $\sim 66\%$ of the time, and wins at *every* bandwidth and var-swap probe tested; the strike-gap mode is a close second, while the pure signed-operator modes do not beat *off* at the median — the reconstruction comes from the tail/strike anchor, not the signed RR/BF operators. The backtest thus *confirmed* the shipped configuration rather than changing a default: the operator bandwidth is not a productive lever (left at 0.06) and the var-swap probe stays at 1.4σ .

8 What is original here

The Bayesian framing (prior precision $\times$ data precision) is standard; the contributions are the *operational* pieces. The *activation gate* (2) turns “do not damp the signal” into one differentiable factor shared by every mode. The *shared precision vocabulary* means strike gaps, operators, factors

and the graph baseline all speak the same language of confidence. The *signed-basket* operators (4.1) keep the RR/BF coupling that per-leg persistence would lose, and incidentally cured the overlay-models’ missing-prior asymmetry. And the *two-pass* option closes the fit-quality circularity exactly. The whole framework is byte-identical to a plain fit when the mode is **off**, so persistence is strictly additive.

A Hyperparameter atlas

Table 1: Prior-persistence hyperparameters.

Knob	Default	Role
<i>Surfaced (OptionsSettings)</i>		
priorPersistenceMode	hybrid	The mode selector (section 4).
priorAnchorWeightPct	50%	Strike-gap budget (fraction of option weight).
priorAnchorDeltas	{2, 5, 10, 25, 40}	Delta-locations for strike anchors.
priorOperatorSet	ATM/RR25/BF25/VS	Operators to persist.
priorOperatorStrengthPct	50%	Operator budget.
priorOperatorRequiredPrecision	1.0	Gate threshold π^{req} .
priorOperatorGapExponent	1.0	Gate sharpness.
γ		
priorOperatorBandwidth	0.06	Operator leg KDE bandwidth.
priorFactorSet	ATM/skew/curv/SMS	Style factors to persist.
priorFactorStrengthPct	50%	Factor budget.
priorTailAnchorStrengthPct	20%	Hybrid deep-tail anchor budget.
priorDataOnlyPrepass	off	Two-pass activation (section 5).
<i>Hidden (precision vocabulary)</i>		
RMS_FLOOR	10^{-4}	Fit-quality precision floor (1 vol bp).
REF_ATM_QUOTES	8	Quote count for full density credit.
SPREAD_HALF	0.05	Relative spread at which precision halves.
OBS/PRIOR_FRESHNESS_HALFLIFE	30 d	Freshness/age half-lives.
TRANSPORT_SCALE	0.10	Transport-distance precision scale.
DEFAULT_BANDWIDTH	0.06	Observed-quote KDE bandwidth.
MAX_INV_VEGA_RATIO	25	Cap on deep-wing vega amplification.
_VARSWAP_PROBE_STD	1.4σ	Var-swap coverage probe half-width.

Performance. The prior blocks add a few residual rows and bump the options version (one refit on change). The single-pass gate is free (it reuses quote-support density); the two-pass prepass costs $\sim 2\times$ per node and is opt-in. Mode **off** is byte-identical to a plain fit.

B Reference implementation: the gated posterior

The gate feeds a precision-weighted Gaussian update. For one coordinate with prior mean μ_0 at (gated) precision λ_0 and a datum d at precision λ_d , the posterior mean is the precision-weighted

average — which the least-squares stack realizes as a residual row $\sqrt{\lambda_0}(\theta - \mu_0)$ appended to the data. The gate multiplying λ_0 is exactly what makes the prior vanish where the data is strong ($\lambda_0 \rightarrow 0 \Rightarrow \text{posterior} \rightarrow d$) and dominate where it is absent ($\lambda_d \rightarrow 0 \Rightarrow \text{posterior} \rightarrow \mu_0$).

Listing 2: The precision-weighted posterior the gated prior row implements.

```

1 def posterior(mu0, lam0, d, lam_d):          # lam0 =
    active_prior_precision (gated)
2     return (lam0*mu0 + lam_d*d) / (lam0 + lam_d)    # -> d as lam0->0;
    -> mu0 as lam_d->0

```

References

- [1] A. Gelman et al. *Bayesian Data Analysis*. CRC Press, 3rd ed., 2013.
- [2] J. Gatheral. *The Volatility Surface*. Wiley, 2006.
- [3] A. Tikhonov and V. Arsenin. *Solutions of Ill-Posed Problems*. Winston, 1977.